

Calculations				
	Sheep	Pig	Cow	Goat
Meat Diet Portion	3.35%	10.82%	10.30%	2.83%
Meat MCal Target	91,623.58	296,095.12	282,045.17	77,573.63
Potential Dairy Value attached to each juvenile	132.54	0.00	1015.20	126.90
Value per juvenile	48.00	160.00	640.00	48.00
Adjusted Dairy Value attached to juvenile based on Dairy Utilization	66.27	0.00	507.60	63.45
Culled Mothers Value	14.77	4.41	105.26	14.77
Culled Adult Males	0.74	0.22	5.26	0.74
Final Value per juvenile	63.51	164.63	750.53	63.51
Value MCals	63.51	164.63	750.53	63.51
Offspring required Year ⁻¹	1,442.72	1,798.59	375.80	1,221.48
MCals from meat	91623.58	296095.12	282045.17	77573.63
Max Utilized Mcals allowed from Dairy	95608.80	0.00	190754.31	77503.16
MCals actually Utilized from Dairy	95608.80	0.00	190754.31	77503.16
Percent of diet from their meat	3.35%	10.82%	10.30%	2.83%
Percent of Diet from their Dairy	3.49%	0.00%	6.97%	2.83%
Kg Fleece	10,902.60	0.00	0.00	0.00
1 Year+ (age) skins	536.36	68.66	72.32	454.16
Yearly Offspring skins	2,625.31	2,119.44	553.79	2,222.83
Birthing Events per year	1.00	1.86	1.00	1.00
Offspring Mother ⁻¹ Year ⁻¹	0.56	5.48	0.68	0.56
Mothers Step 1	2,576.28	328.51	552.64	2,181.22
Males Step 1	721.36	899.29	187.90	610.74
Females Step 1	721.36	899.29	187.90	610.74
Replacement Mothers per year	515.56	66.00	69.27	436.54
Breeding Males Required	135.25	17.25	29.01	114.51
Additional Mothers	2,111.78	58.60	261.76	1,788.12
Males Step 2	591.30	160.43	89.00	500.67
Females 2	591.30	160.43	89.00	500.67
Total Mothers	3,091.83	394.51	621.91	2,617.77
Total Males	1,312.66	1,059.72	276.90	1,111.41
Total Females	130.06	738.87	98.90	110.07
Total Offspring	2,625.31	2,119.44	553.79	2,222.83
Total Animals	4,669.81	2,210.35	1,026.72	3,953.76
Units per Hectare	1.00			
Total Adults per Hectare	1.74	0.27	0.43	1.74
Total Offspring per Hectare	0.86	1.18	0.26	0.86
	2.30%		6.47%	1.95%
Total Adults	5,339	470	913	4,520
Total Offspring	2,625	2,119	554	2,223
Days for Maintenance	5.00	220.00	0.00	5.00
Dry Mothers	466.52	0.00	68.12	394.94
Other	42,005.01	0.00	0.00	35,565.26
MCals Dry Mothers	544,897.40	0.00	144,208.01	461,286.81
total Maintenance MCals	586,902.41	0.00	144,208.01	496,852.06

MCals Required				
	Sheep	Pig	Cow	Goat
Gestating	1,563,374	967,632	927,046	1,323,694
Lactating	2,982,618	343,986	996,823	2,525,355
Maintenance	586,902	0	144,208	496,852
Adult Males	157,977	0	61,422	133,753
Offspring	720,648	2,130,042	2,026,874	610,166
TOTAL	6,011,520.2	3,441,660.0	4,156,372.4	5,089,820.7

Animal Diet Breakdown				
	Sheep	Pig	Cow	Goat
Crop	Barley	Barley	Barley	Barley
Grams ⁻¹ Straw/Chaff	1000.0	0.0	4956.0	1000.0
MCal kg ⁻¹				
Grams ⁻¹ Grain	750.0	750.0	1500.0	750.0
MCal kg ⁻¹				
Days	90.0	90.0	90.0	90.0
Number of Animals	2,625.3	2,119.4	553.8	2,222.8
MCals from Straw/Chaff	335515.0	0.0	350758.1	284077.5
MCals from Grain	531625.9	429187.5	224285.2	450122.8
MCals from other feed	867140.9	429187.5	575043.3	734200.2

Diet Calculations		
Plants Mcal	1,552,383.724	
Meat Mcal	821,250	
Dairy	363,866.27594109	
	Perfect	
Plants %	56.71%	
Meat %	30.00%	
Dairy %	13.29%	
check	Perfect	
Percent		
Domestic Meat	27.30%	
Dairy	13.29%	
Wild Meat	2.70%	
Crops	51.04%	
Wild Plants	5.67%	
check	Perfect	
new dairy	363866.28	
potential dairy < limit	TRUE	
new limit	70.0%	
the limit greater than that left to satisfy	TRUE	
limit	1	
limit dairy checked	FALSE	
Mcalories		Surplus
Domestic Meat	747,337.500	
Dairy	363,866.276	
Wild Meat	73912.500	
Crops	1,397,145.352	
Wild Plants	155,238.372	
check	Perfect	
SURPLUS DAIRY	0.000	
This could account for additional % of diet	0.00%	

Herd Modelling

	Sheep	Pig	Cow	Goat				
Mothers Req'd	3092	395	622	2618				
Males for Slaughter	1313	1060	277	1111				
Females for Slaughter	130	739	99	110				
Offspring available for Consumption	2625	2119	554	2223				
Gestation Req'd	859138	821146	629083	727394				
Lactation Req'd	1639070	291911	676434	1387728				
Maintenance Req'd	1949254	0	185360	1650401				
Adult Males	157977	0	61422	133753				
Offspring Required	862415	2320792	2425603	730199				
Energy Required	5467854	3433848	3977902	4629474				
Ewe Fleece Kg Yr-1	6957							
Ram Fleece Kg year-1	271							
Juvenile Fleece Kg Yr-1	3675							
Kg Fleece Year-1	10903	0	0	0				